DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM
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ALGEBRA (Ph.D. Version)

Instructions to the student

a. Answer all six questions; each will be assigned a grade from 0 to 10.

b. Use a different booklet for each question. Write the problem number and your **code number** (**not** your name) on the outside of the booklet.

c. Keep scratch work on separate pages in the same booklet.

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1. Let G be a group of order $5 \cdot 11 \cdot 17$.

(a) Show that G contains an element of order 187.

(b) Show that if G contains an element of order 55 then G is cyclic.

2. Let M be an $n \times n$ complex matrix and let $R = \mathbb{C}[M]$ be the span over the complex numbers of $I, M, M^2, M^3, \ldots$. Show that M is diagonalizable if and only if the ring R contains no nonzero nilpotent matrices (a nilpotent matrix N is a matrix such that $N^k = 0$ for some positive integer k).

3. Let R be a commutative ring with 1 and let P be a prime ideal of R. Let M be an R-module. Define

$$N = \{m \in M \mid \text{ there exists } r \in R, r \notin P, \text{ with } rm = 0\}.$$

(a) Prove that N is a submodule of M.

(b) Suppose that $N = 0$. Let m be a nonzero element of M. Show that $Pm = \{\pi m \mid \pi \in P\}$ is not equal to Rm.

(c) Suppose that $N = 0$. In addition, assume that M is nonzero and has no proper nonzero submodules. Show that P is a maximal ideal of R.

4. (a) Let R be a principal ideal domain and let M be a finitely generated R-module. Show that there exists an exact sequence

$$0 \to P_1 \to P_0 \to M \to 0,$$

where P_0 and P_1 are projective R-modules.

(b) Let R be a commutative ring with 1 and let M be an R-module. Suppose we have a diagram of R-modules with exact rows

$$\begin{array}{ccccccc} P_1 & \xrightarrow{f_1} & P_0 & \xrightarrow{f_0} & M & \longrightarrow & 0 \\ & & & & \| & & \\ Q_1 & \xrightarrow{g_1} & Q_0 & \xrightarrow{g_0} & M & \longrightarrow & 0 \end{array}$$

where P_0 and P_1 are projective R-modules and Q_0 and Q_1 are arbitrary R-modules. Show that there are homomorphisms $h_0 : P_0 \to Q_0$ and $h_1 : P_1 \to Q_1$ such that the resulting diagram commutes.

5. (a) Let $f(X)$ be a monic irreducible polynomial in $\mathbb{Q}[X]$, and let K be a finite Galois extension of $\mathbb{Q}$. If g, h are monic irreducible factors of f in $K[X]$, show that there exists an automorphism σ of K over $\mathbb{Q}$ such that $g = \sigma(h)$ (applied coefficient-wise).
(b) Give an example where this conclusion is not valid if K is not Galois over $\mathbb{Q}$.

6. Let G be a finite group and let $\rho : G \to \mathrm{GL}_n(\mathbb{C})$ be a representation of G. Let χ be the character of ρ.
(a) Let $g \in G$. Show that $\chi(g^{-1})$ is the complex conjugate of $\chi(g)$. (*HInt:* What does the diagonalization of $\rho(g)$ look like?)
(b) Let $g \in G$. Suppose that g^{-1} is conjugate to g (that is, $g^{-1} = hgh^{-1}$ for some $h \in G$). Show that $\chi(g)$ is a real number.
(c) Let S_m be the group of permutations of m objects. Show that every character of S_m is real valued.